Screen-Printed Flexible Coplanar Waveguide Transmission Lines: Multi-Physics Modeling and Measurement

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***Abstract*— Flexible hybrid electronics (FHE) is a promising technology enabling many applications in biomedical, communication, energy harvesting and internet of things (IoT) areas. To realize FHE applications, the components and devices used in the mentioned technologies need to be electrically characterized under various flexible conditions such as stretching, bending, twisting, and folding. Also, the strain analysis from the mechanical point of view needs to be conducted to justify the reliable applications of FHE under different flexible scenarios. In this paper, the design and electrical characterization of coplanar waveguides (CPWs) in flexible substrates such as Kapton polyimide and polyethylene terephthalate (PET) under uniaxial bending are studied and discussed. The fabricated lines were measured using a vector network analyzer (VNA) up to 8 GHz under both flat and bending conditions. Finite-element models (FEM) of CPW lines were created in ANSYS HFSS to capture the effect of bending on the CPW frequency response. In addition, the variations in the trace width and separations along the CPW lines were modeled accurately to capture the variations in the fabrication process and their effect on the CPW S-parameters in the flat condition. The finite element analysis of strain variation during bending was also performed and the relationship between strain variation and CPW performance was investigated. The bending of the CPW lines was carried out using two parallel plates that had a gap distance varying from 40 mm to 140 mm. The S-parameters were monitored in-situ while the substrate was under bending. The experimental results were compared against simulated results under both flat and bent configurations. Based on the conducted studies, correlation was achieved for the flat and bending scenarios between measurement and simulation results. Also, it was observed that the CPW line has better matching and lower losses compared with the flat case and tensile bending cases.**

Keywords- compressive bending; flexible substrates; screen-printed CPW; strain analysis; tensile bending

I. INTRODUCTION

Flexible hybrid electronics (FHE) including both conventional silicon electronics and printed electronics helps realize many applications in wearable technologies used in biomedical, sensing, energy harvesting and communication areas [1]. For the wearable and implantable electronics used in the healthcare monitoring, the progress in the materials science made it possible to design mechanically compliant electronics [2]. As an example, a microwave CPW transmission line was used as a 40-GHz radio frequency (RF) biosensor in [3] to characterize the dielectric parameters of cancer cells. The various technologies for printing sensors and electronics over flexible substrates including screen-printing and inkjet printing were reviewed and discussed in [4]. Both printing technologies have been widely used to print the microwave devices and components such as antennas, inductors, microstrip and CPW transmission lines on the flexible substrates, for example Kapton and PET.

The RF characterization of CPW transmission lines printed on both rigid and flexible substrates have been reported in several literatures. In [5], the ink-jet printed CPW lines using conductive nano-silver ink on a quartz glass were electrically characterized over the frequency range of 30 kHz to 6 GHz. The derived attenuation from the measured results was reported as 0.15 dB/mm for 10 cm long CPW lines. The authors in [6] have measured and simulated the ink-jet printed CPW lines with silver conductor ink on both flexible Kapton substrates and an Alumina substrate. The line losses of 1.5 dB/mm and 0.8 dB/mm at 60 GHz were reported for the same CPW lines printed on these two substrates, respectively. Also, in [7], a 1mm long CPW line, ink-jet printed on the Kapton substrate, was investigated for high frequency applications up to 67 GHz in both flat and tensile bending conditions. The line losses of 0.3 dB/mm at 67 GHz when sintering at 300℃ was measured in the flat case. The bending of CPW line was performed on a curvature with radii of 15 mm and 25 mm. A minimal variation was observed in the measured attenuation constant of CPW line before and during bending condition. The microwave characterization of ink-jet printed CPW lines on flexible PET substrates over the frequency range of 10 MHz to 20 GHz was explored in [8] and using a space-mapping technique for linking the measured results with simulations, they could extract the conductivity of the ink and dielectric constant of the spacer used during measurements [9]. They also applied their proposed approach on the ink-jet printed CPW lines on Kapton substrates and showed that their methodology can detect small changes in material properties caused by changes in fabrication parameters [10]. The printed electronics over the flexible substrates can be subjected to

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different shape variations such as bending, twisting, stretching and folding, depending on the considered application [1, 2]. Therefore, both electrical and mechanical characterization of FHE under different flexible conditions needs to be carried out to ensure a desirable and reliable performance.

There are some methods for bending test of FHE, such as adaptive curvature flexure, mandrel bending test, etc. The main advantage of adaptive curvature flexure over mandrel bending test is that the flexible substrate can be subjected to different radii of curvature through moving parallel plates using just one test setup. The distance between two parallel plates, called panel separation (PS), can be adjusted for having different radii of curvature [11]. Using this bending test, an aerosol jet printed (AJP) microstrip transmission line was characterized both electrically and mechanically through finite element modeling in the tensile and compressive bending tests and the measured results were compared against simulation. It was shown that there is little change in DC resistance of the transmission line during bending and the change in either return loss or insertion loss at different panel separations (40 mm – 100 mm) compared to the flat condition was not noticeable [12].

Compared to previous work [5-10], this paper presents three main contributions on screen-printed CPW lines over flexible substrates such as Kapton and PET. First, the effect of inherent variability involved in the fabrication process on the CPW frequency response was investigated through line profile measurements and multi-section modeling of the CPW line in ANSYS HFSS [13]. Second, as opposed to [7], the CPW line with a length of 166.37 mm was measured under different bending (tensile and compressive) scenarios using adaptive curvature bending for a wide range of panel separations (40 mm – 140 mm) and the results were then correlated with measurements using ANSYS HFSS. Third, the strain variations during bending were studied using a 2D finite-element model of CPW line in ANSYS [14] and its impact on the electrical properties of the CPW line was explored. To the best of our knowledge, the above-mentioned contributions have not been reported in the open literature.

This paper is organized as follows. In section II, the dimensions of CPW lines screen-printed on Kapton and PET substrates are presented. The electrical characterization of CPW lines including measurement and simulation together with the impact of process variations on CPW frequency response (flat case) are described in section III. Section IV presents the measurement and simulation results of CPW lines in both tensile and compressive bending cases along with the analysis of strain variations during bending conditions. Finally, the concluding remarks are given in section V.

II. Dimensions of CPW Lines

The CPW lines were designed and screen-printed at Georgia Tech and DuPont, respectively. Figure 1 shows the fabricated CPW lines (L1 and L2) along with TRL (Through-Reflect-Line) structures on a 5 mil Kapton and PET substrates. The DuPont 5028 silver conductor [15] was used to print both the signal (trace) and ground planes. All the results presented in this paper are related to the longest CPW line, L2, with a length of 166.37 mm. The fabricated TRL structures (L3, L4, and Open/Short), used in the de-embedding process to remove the effect of SMA connectors from the measurement results, did not work as expected. Therefore, the de-embedding of measurement results presented later was not implemented.

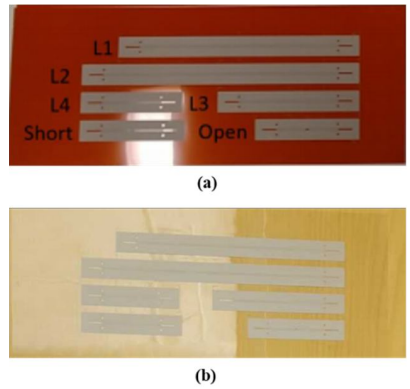


Figure 1. The screen-printed CPW lines on (a) Kapton and (b) PET substrates.

A cross-sectional view and a top view of the CPW line is depicted in Fig. 2. As shown, the CPW line consists of a center conductor (trace) with two finite-dimension ground planes on each side of a slot (separation or gap).

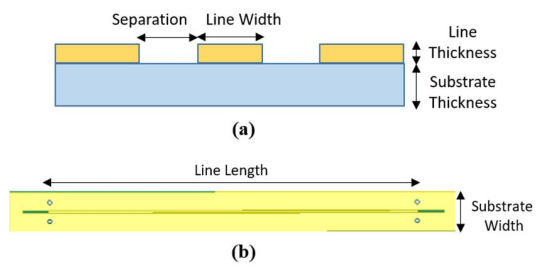


Figure 2. The cross-section (a) and the top view (b) of the CPW lines.

The dimensions of CPW lines are listed in Table I. The line width and separations were measured using an optical microscope while cross-section dimensions such as the thickness and surface roughness were measured using a Veeco Dektak 150 surface profilemeter. Because of variations in the line width and separations along the CPW lines after fabrication, the average values for these parameters are presented in Table I. To determine the conductivity of fabricated CPW lines, the DC resistance of the lines was measured using a digital multimeter (DMM). Then, the conductivity was calculated using $\sigma = L/(R.A)$ where σ is the line conductivity, L is the line length, R is the measured DC resistance, and A is the cross-sectional area of the CPW line. It is worth noting that the cross-sectional area of CPW lines is assumed as a rectangular area which equals the product of the line width and the line thickness as listed in Table I. This assumption is used in the transmission line modeling (for simplification) as mentioned in [5-10, 12].

However, such cross-sectional area does not typically occur in the fabricated CPW lines and in practice, it is a trapezoidal cross-section as shown in Fig. 3.

TABLE I. DIMENSIONS OF CPW LINES

CPW Parameters	Value	
	Substrate: Kapton	***Substrate: PET***
Substrate length (mm)	196.418	196.418
Substrate width (mm)	14.82	14.82
Line length (mm)	166.37	166.37
Line width (µm)	762.07	764
Line thickness (µm)	10	10
Separation (µm)	106.68	90.3
Surface roughness (µm)	1	1
Line conductivity (S/m)	$2.974*10^6$	$2.9*10^6$

The calculated conductivities presented in Table I are estimations of the actual conductivities of CPW lines which have a trapezoidal cross-section. These conductivities are used for CPW line simulations in ANSYS HFSS as a constant value for the entire frequency range because the average values of the line width and thickness were used in the calculations.

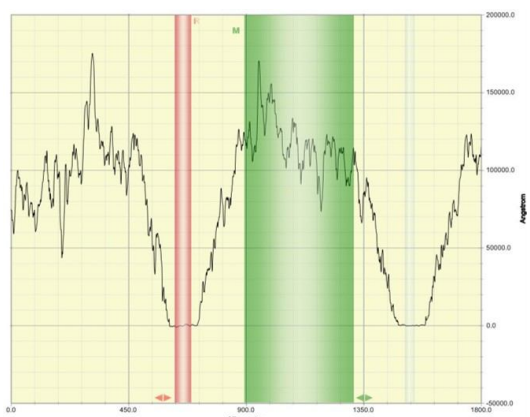


Figure 3. Measured trace profile of the CPW line with Kapton substrate.

The relative permittivity and the loss tangent of the Kapton substrate are 3.5 and 0.0026 at 1 kHz [16], respectively, while these values for the PET substrate characterized using the ring resonator method in the frequency range of 0 to 6 GHz are 2.9 [17] and 0.0314 [18], respectively.

III. MEASUREMENT AND MODELING OF CPW LINES (FLAT CASE)

A. Measurement Results

To measure the S-parameters of CPW lines, an Agilent E8363B VNA was used and calibrated using an HP 85052D Calibration Kit where SOLT (Short-Open-Load-Through) calibration was used to calibrate until the edge of the SMA connectors. The VNA cables were connected to each side of CPW lines using a VLF40-002 SMA which is a vertical launch connector. The CPW lines with mounted SMAs are depicted in Fig. 4.

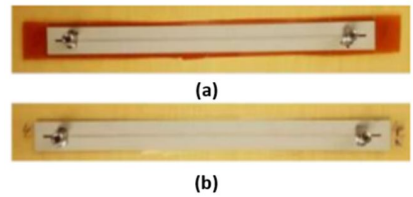


Figure 4. CPW lines with mounted SMAs screen-printed on (a) Kapton and (b) PET substrates.

Figure 5 shows the measured S-parameters of CPW lines in the flat condition while floating in the air in the frequency range of 10 MHz to 8 GHz. As seen, the CPW line printed on the Kapton substrate is less lossy than the CPW line printed on the PET substrate. This is because of the higher loss tangent of PET substrate compared to that of Kapton substrate which has a great impact at higher frequencies. Also, the line conductivity printed on PET substrate is a little bit smaller than that printed on the Kapton substrate. The measured losses for the CPW lines printed on the PET and Kapton substrates are 0.09057 dB/mm and 0.0749 dB/mm at 8 GHz, respectively.

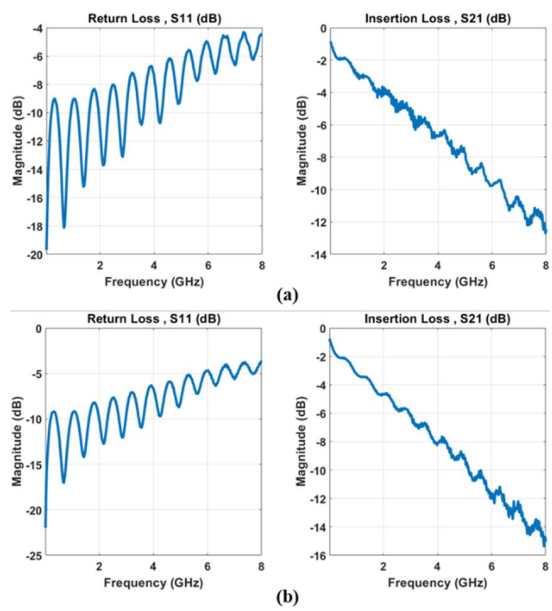


Figure 5. Measured return loss (S_{11}) and insertion loss (S_{21}) of CPW lines screen-printed on (a) Kapton and (b) PET substrates in the flat case.

The skin depth of the printed silver ink on the Kapton and PET substrates at 8 GHz were calculated as 3.263 µm and 3.304 µm, respectively, which are more than the measured surface roughness listed in Table I. Therefore, the surface roughness doesn't have a significant impact on the attenuation of CPW lines. The measured return loss for both lines is above -10 dB approximately in the whole frequency range which shows a high impedance mismatching for these lines.

B. *Simulation and Measurement Correlation*

The finite element models of CPW lines were created in ANSYS HFSS [13] using the parameters listed in Table I and the material properties of both substrates mentioned earlier. The simulation results were compared against the measured ones for both CPW lines as shown in Fig. 6. A good agreement between the simulated and measured results is observed for both CPW lines. The discrepancies seen in the comparison of insertion loss results may be due to the conductivity value used in the simulation models which is an estimate for the actual conductivity value of the lines, as explained in section II. The conductivity values listed in Table I were calculated with the assumption of the rectangular cross-section of the line which is a simplification for the trapezoidal cross-section as shown in Fig. 3. In addition, the process variations contribute to these discrepancies which are discussed in the following sections.

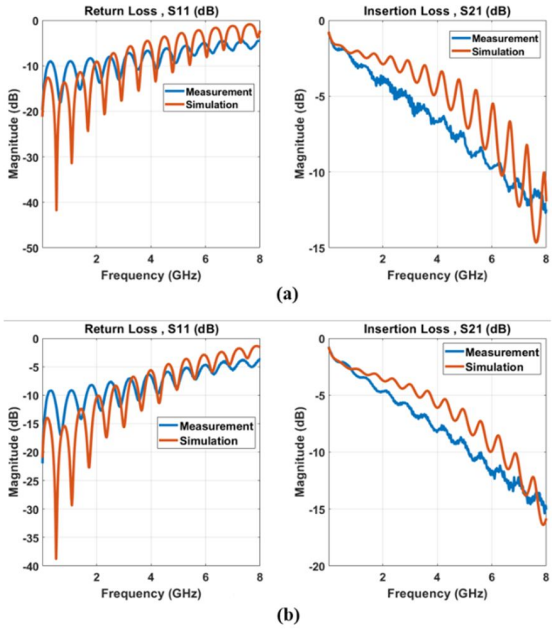


Figure 6. Correlation between the measurement and simulation results for CPW lines printed on (a) Kapton and (b) PET substrates in the flat case.

C. *Study of Fabrication Process Variations*

The process variations in the fabricated CPW lines appear in the form of variations in the line width and separations along the lines. To demonstrate the effect of process variations on the CPW S-parameters, we measured the profile of the CPW line printed on the PET substrate at 20 equal-length sections as shown in Fig. 7, in which sections 1 and 20 represent the leftmost side and the rightmost side of the CPW line. The maximum change in the line widths among 20 sections is 107.1 µm which is 14% of the average line width listed in Table I. As shown in Fig. 2 (a), there are two separations along the CPW lines while we listed one single value for both separations in Table I. It means that we assumed a symmetric structure for the CPW line which is not valid in practice. From the 20 measured profiles, the maximum change in the left and the right separations are 106.66 µm and 108 µm, respectively, which are 118.12% and 119.6% of the average separation value listed in Table I, respectively. As we traverse along the line from section 1 to section 20, the trend of measured data points for both separations is increasing while that for the line width is decreasing.

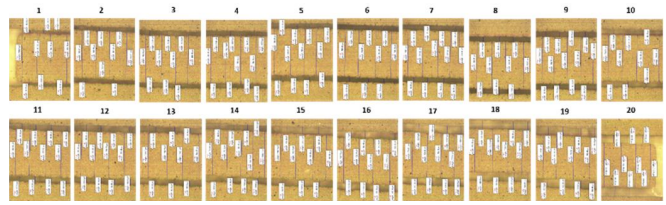


Figure 7. Measured profiles of CPW line printed on PET substrate at 20 equal-length sections.

The finite element model of the CPW line with 20 measured sections was created in ANSYS HFSS and the obtained results for the CPW line printed on the PET substrate were compared against the results shown in Fig. 6 (b). As seen, a noticeable improvement in the correlation of the insertion loss is observed compared to the simulation result without considering the 20-section profile which is expected. As the number of sections of the CPW profile measurements increases, we get closer to the actual geometry of the CPW line. As a result, a better correlation is achieved between the simulation and measurement results. Because the characteristic impedance of the CPW lines mainly depends on the line width and separation, the variations in the line width and separation along the line cause impedance variation or mismatch which results in more reflections along the line as shown in Fig. 8.

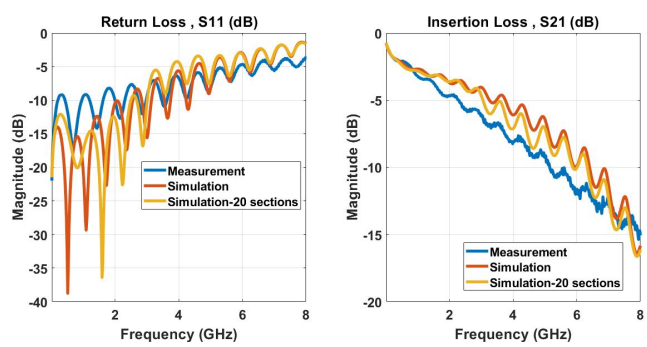


Figure 8. Comparison of simulation results from 20-section CPW line printed on PET substrate with the results of Fig. 6 (b).

IV. Measurement and Modeling of CPW Lines for Bending Cases

In this section, the bending measurements performed on the CPW line printed on PET substrate are only presented and discussed as the same trend for bending measurement results was observed for the CPW line printed on Kapton substrate. After performing SOLT calibration on the VNA, the bending measurements of the CPW line were carried out using the adaptive curvature bending test setup shown in Fig. 9. The

CPW line is placed between two plates where the bottom plate is fixed while the top plate is moved up and down to have different panel separations. There are two flat sections on either side of the CPW line which are where the VNA cables are connected to the SMAs via a cutout in the plate. The laboratory setup for both tensile and compressive bending measurements of the CPW line is shown in Fig. 10.

The plates made of high-density polyethylene (HDPE) material have a relative permittivity of 2.3 and loss tangent of 0.0005 [13]. Under tensile bending, the CPW line is on the convex side (outside) surface of the substrate, while under compressive bending, the CPW line is on the concave side (inside) surface of the substrate.

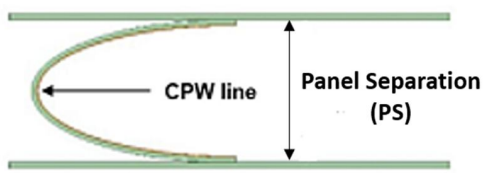
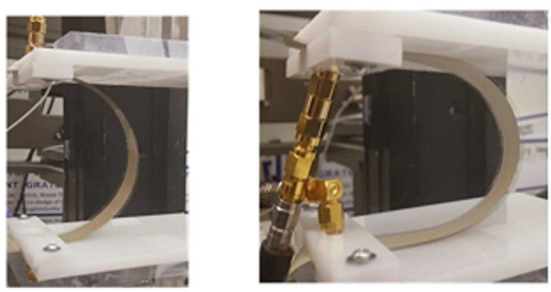


Figure 9. Illustration of the adaptive curvature bending test setup.

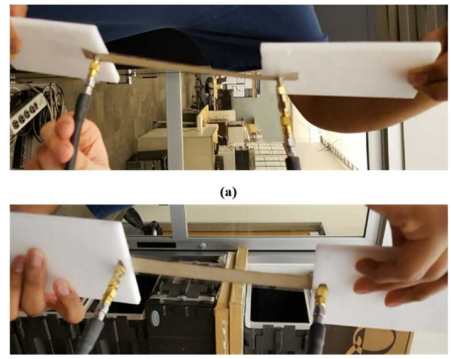


(a) (b)

Figure 10. The laboratory setup for the CPW line (a) tensile and (b) compressive bending measurements.

Both bending measurements were carried out at the panel separations of 40 mm to 140 mm. These measurements were compared against the flat measurement result of the CPW line to observe the effect of bending on the CPW frequency response. To have an accurate comparison, two flat measurements of the CPW line, as shown in Fig. 11, were also performed in which small parts from both ends of the CPW line were attached to the HDPE panels and the remaining part of the line is floating in the air.

These two flat measurement results were compared with those shown in Fig. 5 (b) in which the CPW line was freely floating in the air without those panels. It shows that there is a very negligible difference between these three cases such that the results of Fig. 5 (b) can be considered as the reference case to compare with bending measurement results.



(a)

(b)

Figure 11. Two flat measurements of the CPW line using the HDPE panels.

A. Measurement Results

1) Tensile Bending: The comparison of tensile bending measurement results of the CPW line with the flat case is shown in Fig. 12. As seen, the frequency response of the CPW line changes while undergoing tensile bending. Although, the return loss is not better than 10 dB (matching criterion) for the flat case and for all bending cases over the whole frequency range, it is observed that the return loss of all bending cases is better than that of the flat case, indicating a better match.

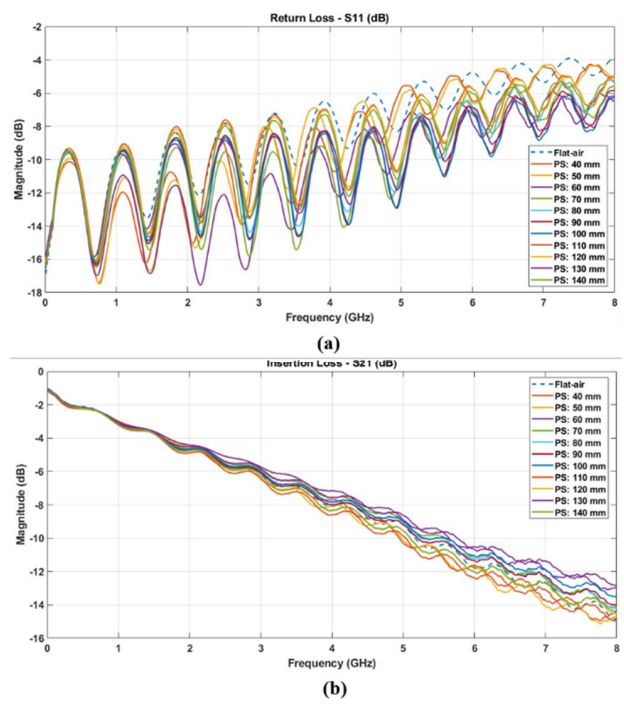


Figure 12. Comparison of flat and tensile bending measurement results for the CPW line, (a) Return loss, (b) Insertion loss.

To have a better view for the change in insertion loss during bending, the magnitudes of insertion loss for all bending cases were normalized to that of the flat case (-14.93

dB) at 8 GHz as shown in Fig. 13. As seen, the maximum and the minimum change in the insertion loss occur at the panel separation of 110 mm and 130 mm, respectively. In other words, the maximum and the minimum deviation in the insertion loss compared to the flat case at 8 GHz are 2.05 dB and 0.09 dB, respectively. Also, the range of variation in the insertion loss among all bending cases is 2.14 dB at 8 GHz. From Fig. 12, it seems that no clear relationship can be found for the change in results for different panel separations.

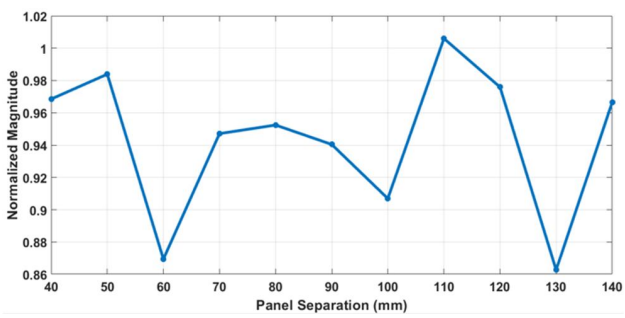


Figure 13. Normalized magnitude of insertion loss for all tensile bending cases with respect to the flat case at 8 GHz.

2) Compressive Bending: Fig. 14 compares the compressive bending measurement results of the CPW line with the flat case. As seen, there is a small difference between the results of different panel separations. Also, the measured return loss of all compressive bending cases compared with the flat case is better than 10 dB over the entire frequency range.

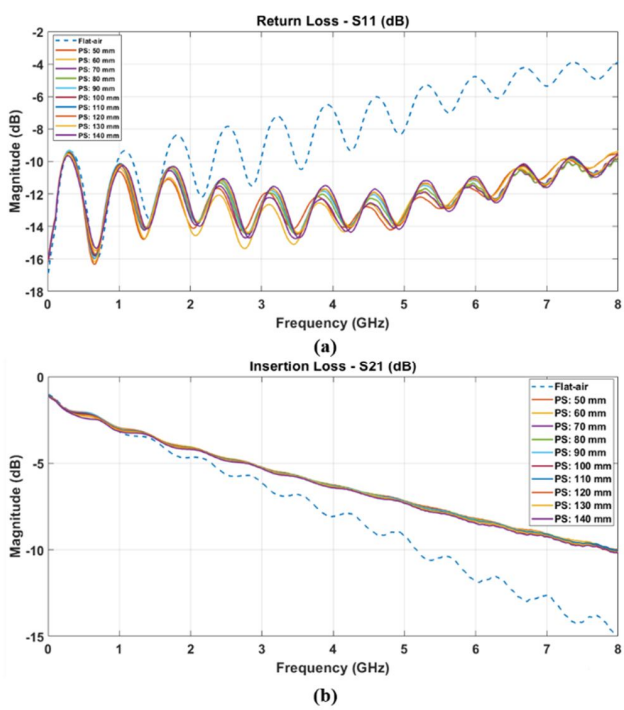


Figure 14. Comparison of flat and compressive bending measurement results for the CPW line, (a) Return loss, (b) Insertion loss.

However, the maximum deviation in the insertion loss between the flat case and all compressive bending cases is around 4.93 dB at 8 GHz which shows 33% improvement in the insertion loss. The reason behind the better insertion loss in all compressive cases is not clear yet and needs to be investigated further. Comparing Fig. 12 and Fig. 14 shows that the CPW line under compressive bending exhibits better matching and lower insertion loss compared with the flat case and all tensile bending cases as well.

B. Study of Strain Variations for the Bending Cases

To better understand the effect of bending on the electrical properties of the CPW line, a 2D mechanical finite element analysis of strain variations during tensile bending was carried out in ANSYS 18.1 [14], as shown in Fig. 15.

From these simulations, it was found that the change in the average relative resistance during bending is very small such that the line conductivity under flat condition can be used for the tensile and compressive bending simulations in HFSS as well. This finding was also observed in [12, 19] for the microstrip transmission line under bending scenarios.

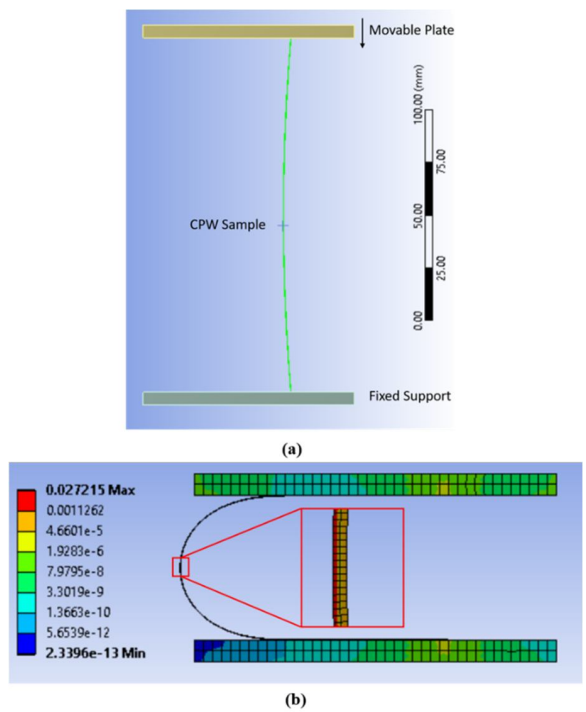


Figure 15. The 2D model for mechanical finite-element analysis of CPW line. (a) the simulation setup. (b) maximum principal strain distribution at the panel separation of 40 mm while CPW line under tensile bending.

The relative resistance (R/R_0) is calculated using (1) as follows [19]:

$$\frac{R}{R_0} = \int_0^L \frac{(1+\epsilon(s))^2 \rho(\epsilon(s))}{L} ds. \quad (1)$$

Where L is the length of the line, ρ(ε) is the relationship between resistivity and tensile strain, and ε(s) is the tensile

strain at position s along the CPW line. The function $\rho(\varepsilon)$ is the same as that used in [19] for the screen-printed microstrip line which is $\rho = \rho_0(1+20.5\ \varepsilon)$.

The relationship between resistivity and compressive strain is not determined yet and needs to be investigated further in the future work. However, as the strain under compressive bending is less than that under tensile bending, the change in relative resistance would be smaller than that under tensile bending [12].

Table II presents the finite element analysis results of the average strain, the average relative resistance and the maximum strain obtained at the panel separations of 40 mm to 140 mm for tensile bending of the CPW line. As we traverse along the CPW line, the maximum strain occurs at the middle point of the line for panel separations of 40 mm-100 mm, while for those of 110 mm-140 mm, the maximum strain occurs at the clamps (SMA locations at both ends). The maximum increase in the relative resistance is 2.43% which occurs at the panel separation of 40 mm. It corresponds to a relative conductivity of 0.9763 which shows a relative conductivity decrease of 2.37%. Therefore, the conductivity of the CPW line (L2) for tensile bending at this panel separation reduces to 2.83e6 S/m as compared to 2.9e6 S/m under flat conditions. In the following subsection, the tensile and compressive bending simulation results are presented and compared with the corresponding measurement cases.

TABLE II. ESTIMATED CHANGE IN AVERAGE RELATIVE RESISTANCE FOR CPW LINE UNDER TENSILE BENDING

Panel Separation (mm)	Average Strain	Average Relative Resistance	Maximum Strain
40	0.001184196	1.024276026	0.0037
50	0.001172991	1.024046307	0.003
60	0.0011623	1.023827153	0.0025
70	0.001151256	1.023600754	0.0021
80	0.001145209	1.023476784	0.0018
90	0.001144059	1.023453217	0.0016
100	0.001141791	1.023406712	0.0013
110	0.001139223	1.023354081	0.0014
120	0.001136177	1.02329162	0.0019
130	0.001132146	1.023208994	0.0027
140	0.001125645	1.02307572	0.0039

C. Simulation and Measurement Correlation

The adaptive curvature bending setup for both tensile and compressive bending simulations was modeled in ANSYS HFSS accurately to capture the conditions of the measured CPW line in the laboratory. Figure 16 shows the HFSS models for both tensile and compressive bending simulations of the CPW line. The HFSS simulations were carried out in the frequency range of 10 MHz-8 GHz with 401 points for interpolating frequency sweep at the mentioned range of the panel separation. The correlation of simulation results against the measurement ones is only discussed for three of panel separations which are 60 mm, 90 mm and 120 mm for both bending cases. The same discussions can be applied to the results for other panel separations as well.

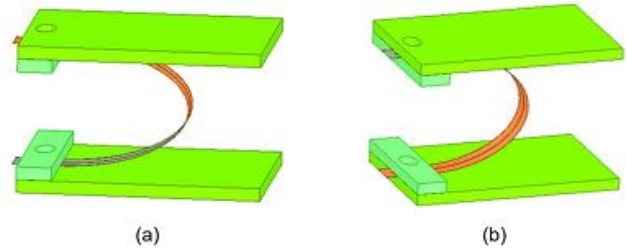


Figure 16. HFSS models for (a) tensile and (b) compressive bending simulations of the CPW line.

1) Tensile Bending: The comparison of simulation and measurement results at panel separations of 60 mm, 90 mm and 120 mm is presented in Fig. 17. Good agreement between the simulation and measurement results of the insertion loss is observed for all panel separations shown in Fig. 17. Also, for the return loss, good correlation is achieved between the simulation and measurement results by 3 GHz where some small deviation between the results start appearing and continues up to 8 GHz.

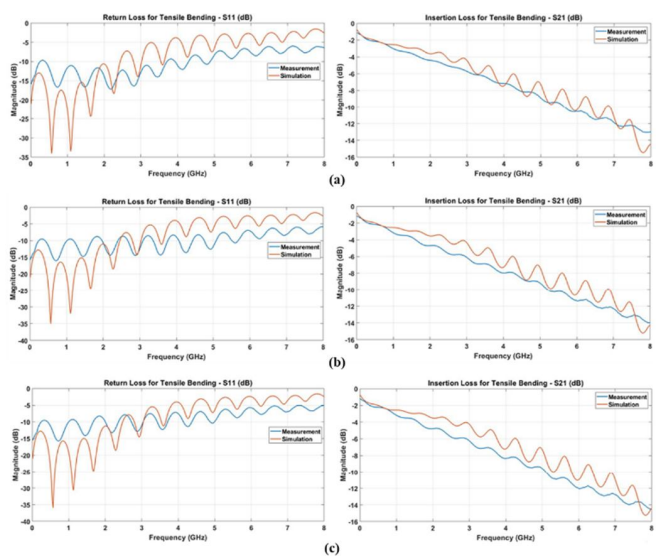


Figure 17. Correlation between the measurement and simulation results for tensile bending of the CPW line at panel separations of (a) 60 mm, (b) 90 mm, and (c) 120 mm.

2) Compressive Bending: The comparison of simulation and measurement results at panel separations of 60 mm, 90 mm and 120 mm is shown in Fig. 18. Once again, good agreement between simulation and measurement results for

insertion loss is observed for all panel separations up to 6 GHz. Beyond this frequency, the simulation results start deviating from measurements such that the deviation in results reaches around 5 dB at 8 GHz. One reason for this deviation could be that HDPE panels at both ends of the CPW line for compressive bending behave as another layer of dielectric attached to the PET substrate. This added layer appears to have a larger impact at higher frequencies in the simulation results. For return loss, a good correlation between simulation and measurement results was achieved for all panel separations up to 3 GHz. But, the simulation results start diverging from measurements after 3 GHz such that the resulting discrepancy reaches around 7 dB at 8 GHz. The process variations along the CPW line discussed in section III can be another reason for the result discrepancies at higher frequencies.

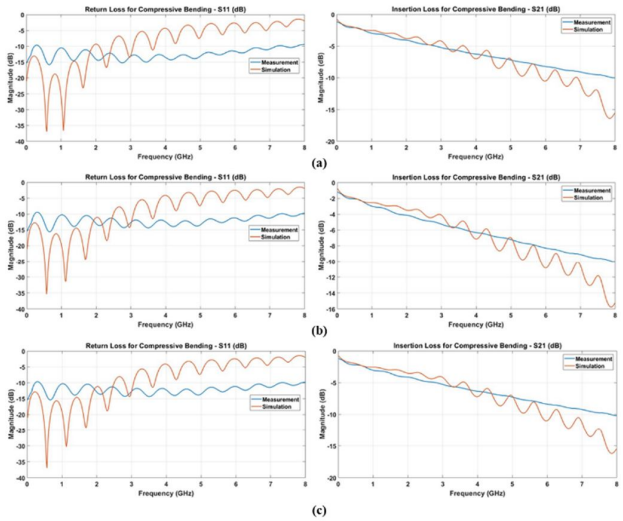


Figure 18. Correlation between the measurement and simulation results for compressive bending of the CPW line at panel separations of (a) 60 mm, (b) 90 mm, and (c) 120 mm.

V. Conclusion

The CPW lines screen-printed on two flexible substrates, Kapton polyimide and PET were electrically characterized and the measurement results were compared against simulation results from the FEM models created in ANSYS HFSS. The insertion loss of CPW lines printed on PET and Kapton substrates was measured as 0.09057 dB/mm and 0.0749 dB/mm at 8 GHz, respectively. However, the measured insertion loss for the AJP microstrip line [12] and screen-printed microstrip line [19], both printed on Kapton polyimide substrate, was reported as 0.125 dB/mm and 0.04 dB/mm at 8 GHz, respectively. The simulation results agreed reasonably well with measurements in the flat case. The effect of process variations on the CPW frequency response was investigated through multi-section modeling of the CPW line in HFSS. Simulation results verified the impact of process variations which exhibits itself in the form of variations in the line width and separation. By increasing the number of sections, improvement in the correlation of insertion loss was expected. Also, both tensile and compressive bending measurements and simulations of the CPW line using the adaptive curvature bending were performed at different panel separations ranging from 40 mm to 140 mm and the analysis of the strain variations during bending cases showed that there is a little change in the average relative resistivity of the CPW line. The discrepancies observed in the simulation to measurement correlation for all bending cases showed that there is still room for improvement which needs to be investigated further in future work. Compressive bending measurements of the CPW line demonstrated that the CPW line has better matching and lower losses compared with the flat case and tensile bending cases.

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